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1**Ion distribution and electrochemical gradients**

COMP

You should be able to: State the typical intracellular and extracellular concentrations of sodium, potassium, chloride, and calcium and separate the chemical from the electrical component of the driving force on each ion.

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2**Nernst equation**

COMP

You should be able to: Calculate the equilibrium potential for an ion with the Nernst equation and explain what the sign of the result means for the direction that ion will move.

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3**Resting membrane potential**

COMP

You should be able to: Explain why the resting membrane potential sits near the potassium equilibrium potential and predict how it shifts when membrane permeability to potassium or sodium changes.

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4**Ion channel gating**

COMP

You should be able to: Compare leak, voltage gated, ligand gated, and mechanically gated channels by what opens each one and give a physiological location for each.

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5**Depolarization and hyperpolarization**

COMP

You should be able to: Define depolarization, repolarization, hyperpolarization, and overshoot and label each on a membrane potential tracing.

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6**Membrane potential simulation**

COMP

You should be able to: Manipulate extracellular potassium and sodium in a simulation and record the resulting change in resting membrane potential against the Nernst prediction.

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7

Neuron structural and functional classes

COMP

You should be able to: Classify a neuron as multipolar, bipolar, or pseudounipolar by structure and as sensory, motor, or interneuron by function, and match each class to a location in the nervous system.

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8

Functional regions of a neuron

COMP

You should be able to: Label the dendrites, cell body, axon hillock, trigger zone, axon, and axon terminal and state which signal type each region carries.

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9**Glial cell functions**

COMP

You should be able to: Match astrocytes, oligodendrocytes, microglia, ependymal cells, satellite cells, and Schwann cells to their physiological roles.

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10**Myelin and its loss**

COMP

You should be able to: Explain how myelin and the nodes of Ranvier speed conduction and predict the functional consequence of demyelination.

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11**Axonal transport**

COMP

You should be able to: Distinguish fast anterograde and retrograde axonal transport from slow transport and state what each carries.

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12**Graded potentials**

COMP

You should be able to: Describe how a graded potential is produced and explain why it varies with stimulus strength and decays with distance, using the terms current leak and cytoplasmic resistance.

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13**Action potential phases**

COMP

You should be able to: Diagram an action potential and state the channel state and ion movement responsible for depolarization to peak and for repolarization and afterhyperpolarization.

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14**Threshold and all or none**

COMP

You should be able to: Explain what threshold represents at the trigger zone and why action potential amplitude does not change with a stronger stimulus.

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15**Coding of stimulus intensity**

COMP

You should be able to: Explain how the nervous system encodes stimulus strength when action potential amplitude is fixed.

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16**Refractory periods**

COMP

You should be able to: Distinguish the absolute from the relative refractory period by channel state and explain how the refractory period sets maximum firing frequency and prevents backward conduction.

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17**Conduction velocity**

COMP

You should be able to: Rank axons by conduction velocity using diameter and myelination and contrast continuous with saltatory conduction.

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18**Effects of altered extracellular ions**

COMP

You should be able to: Predict the effect of hyperkalemia and hypokalemia on resting potential and excitability and explain why local anesthetics that block sodium channels abolish the action potential.

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19**Action potential simulation**

COMP

You should be able to: Run a neuron simulation with sodium and potassium channel blockers and interpret the resulting change in the action potential tracing.

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20**Nerve conduction measurement**

COMP

You should be able to: Measure conduction velocity from a recorded compound action potential and account for the difference between the fastest and slowest fibers.

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21**Sequence at a chemical synapse**

COMP

You should be able to: Order the events of chemical synaptic transmission from action potential arrival through calcium entry and vesicle fusion to receptor binding and postsynaptic response.

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22**Neurotransmitter classes**

COMP

You should be able to: Match acetylcholine, the catecholamines, serotonin, glutamate, GABA, glycine, and the neuropeptides to their usual excitatory or inhibitory effect and to a site of action.

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23**Neurotransmitter removal**

COMP

You should be able to: Name the three routes that clear a neurotransmitter from the synaptic cleft, degradation, reuptake, and diffusion, and give a drug that blocks one of them.

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24**Excitatory and inhibitory postsynaptic potentials**

COMP

You should be able to: Distinguish an EPSP from an IPSP by the ion channel opened and the direction of the membrane potential change.

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25**Summation and integration**

COMP

You should be able to: Distinguish temporal from spatial summation and determine whether a stated combination of EPSPs and IPSPs will bring the trigger zone to threshold.

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26**Presynaptic modulation**

COMP

You should be able to: Compare presynaptic inhibition and facilitation with postsynaptic modulation and state the advantage of presynaptic control at a single input, which is selectivity.

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27**Synaptic plasticity**

COMP

You should be able to: Explain long term potentiation as a mechanism of learning and identify the roles of repeated stimulation and receptor insertion.

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28**Electrical synapses**

COMP

You should be able to: Compare an electrical synapse at a gap junction with a chemical synapse by speed and by capacity for modulation.

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29**Synapse simulation**

COMP

You should be able to: Alter neurotransmitter release and receptor availability in a synapse simulation and interpret the resulting postsynaptic recording.

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Dr. Sharilyn Rennie